

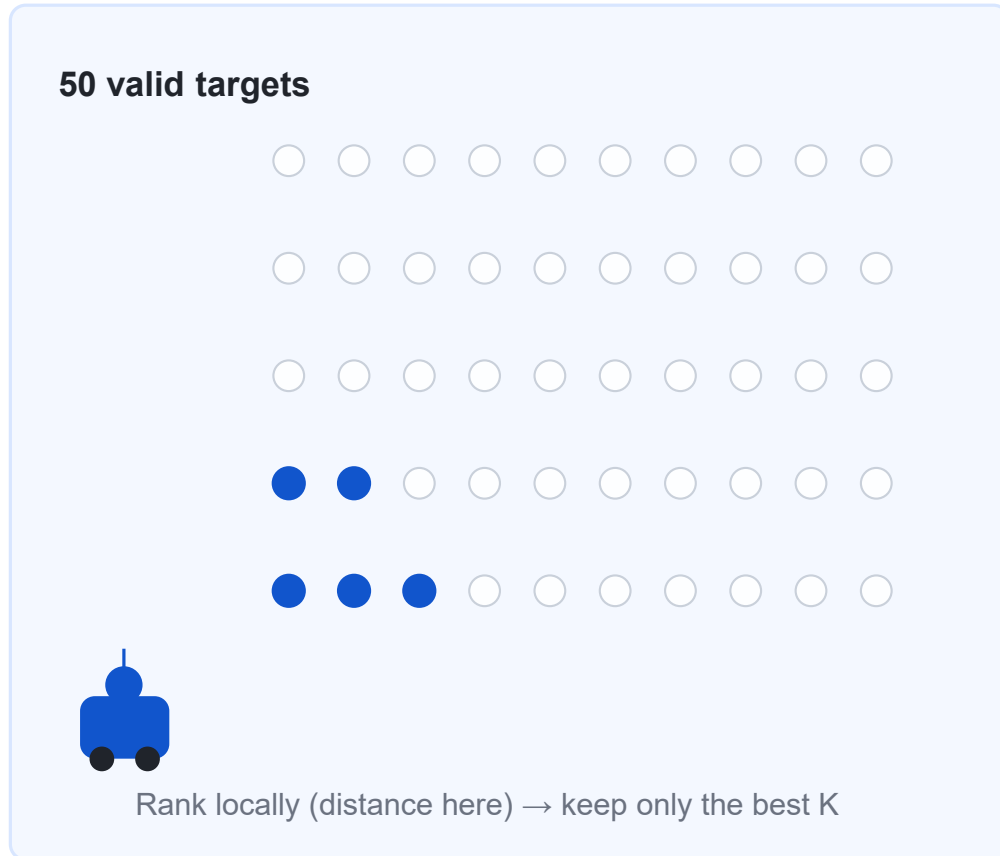
Candidate-Set Restriction for Computationally Efficient Decentralized Multi-Robot Task Allocation

Presenter: James Lott
Advisor: Dr. Vahraz Honary

MIT IEEE URTC 2026



Do robots need to consider every task?



Same allocator logic. Smaller problem.

4
robots

50
targets

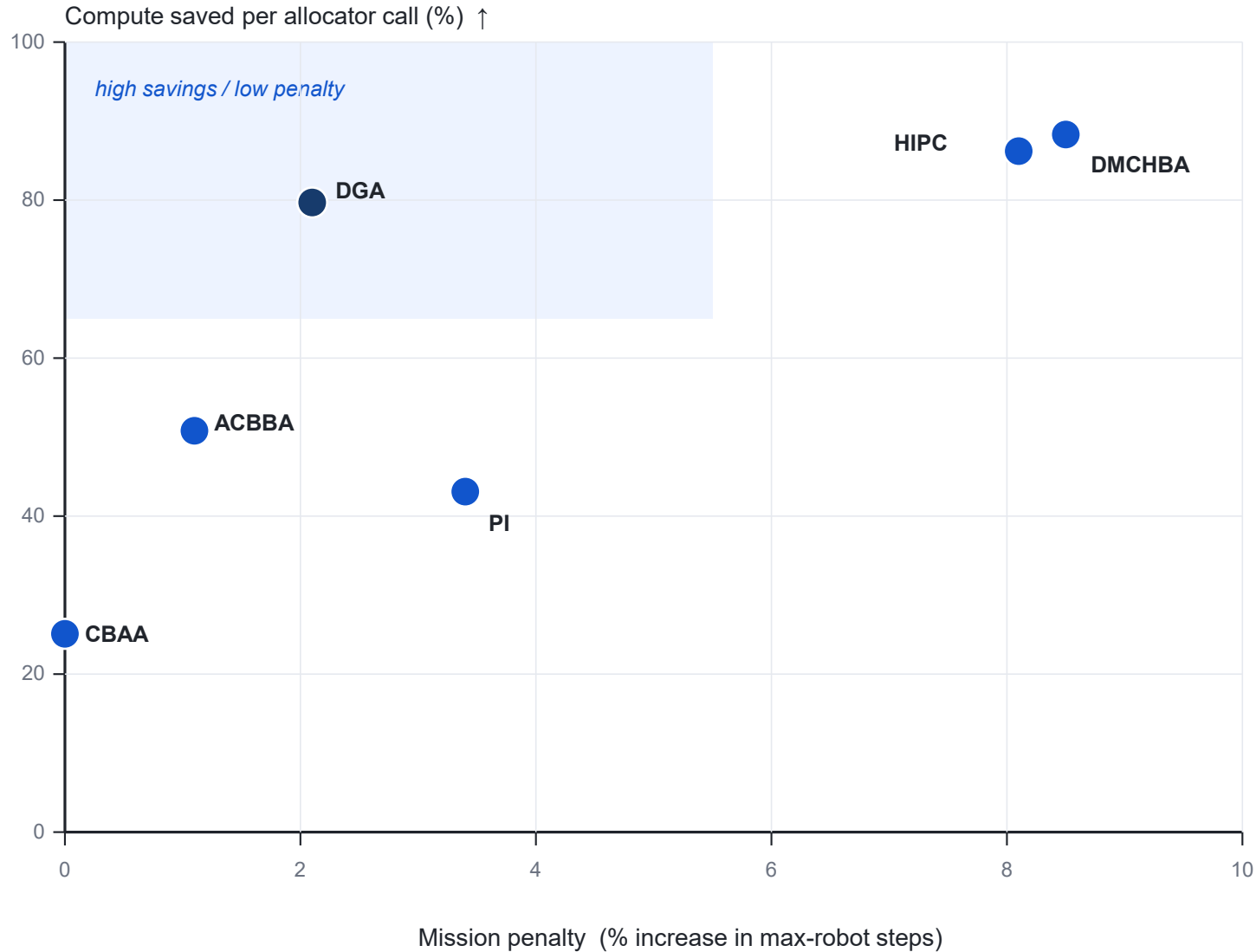
6
allocators

8
K levels

4,800
paired trials

At 10% retention, compute fell faster than performance

K = 5 of the initial 50 targets — paired change relative to unrestricted execution.



62.2%

mean compute reduction

filtering overhead included

3.9%

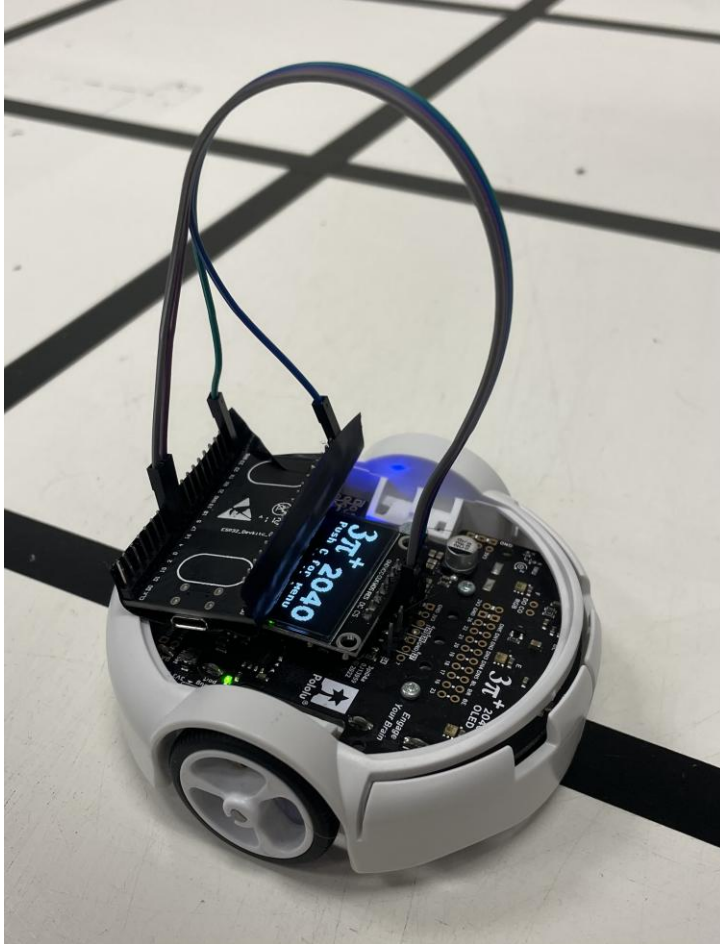
mean max-step degradation

response is allocator-specific

Example: DGA saved 79.7% compute with +2.1% steps.

On the RP2040, restriction changed what was runnable

For DMCHBA, candidate restriction crossed an embedded execution-feasibility boundary.



Pololu 3pi+ 2040 • RP2040 microcontroller

DMCHBA on RP2040

100%	K = 50	> 30 s guard	×
75%	K = 38	> 30 s guard	×
50%	K = 25	feasible	✓
10%	K = 5	≈ 96 ms / call	✓

Candidate-set size is a deployment parameter.